
Optical Fiber Propagation
Main Features
Chap 2, Agrawal, 4th Edition



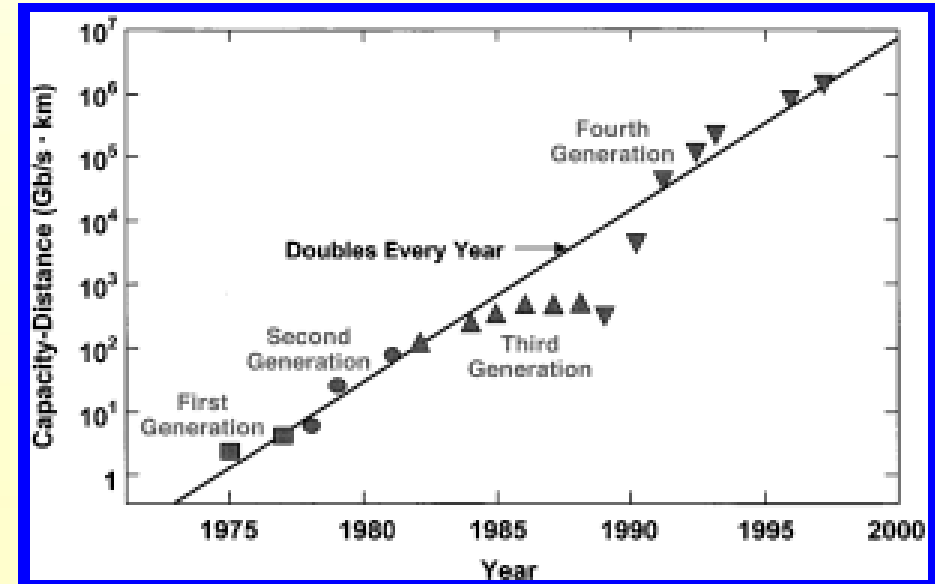
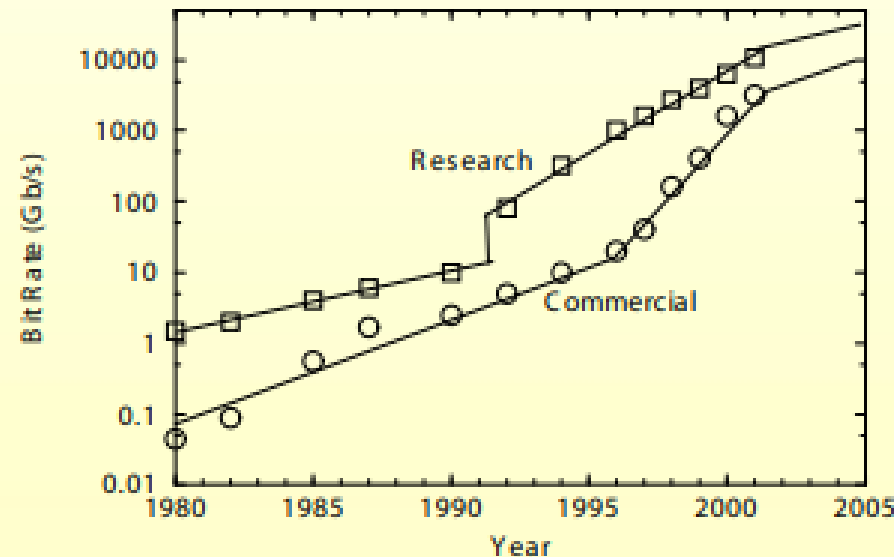
Information Revolution

- Industrial revolution of 19th century gave way to information revolution during the 1990s.
- Fiber-Optic Revolution is a natural consequence of the Internet growth.



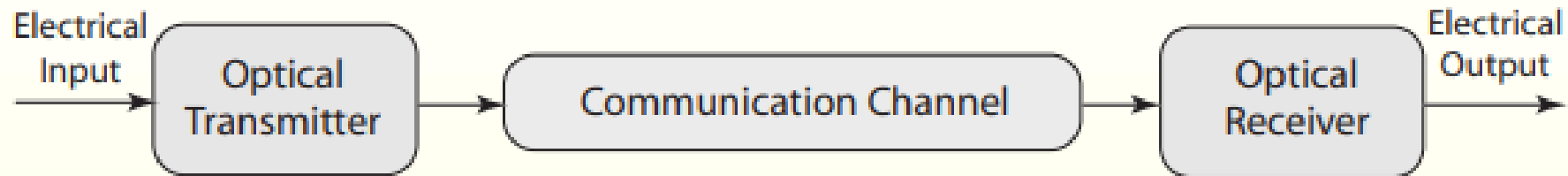
Five Generations

- 0.8- μm systems (1980); Graded-index fibers
- 1.3- μm systems (1985); Single-mode fibers
- 1.55- μm systems (1990); Single-mode lasers
- WDM systems (1996); Optical amplifiers
- L and S bands (2001); Raman amplification

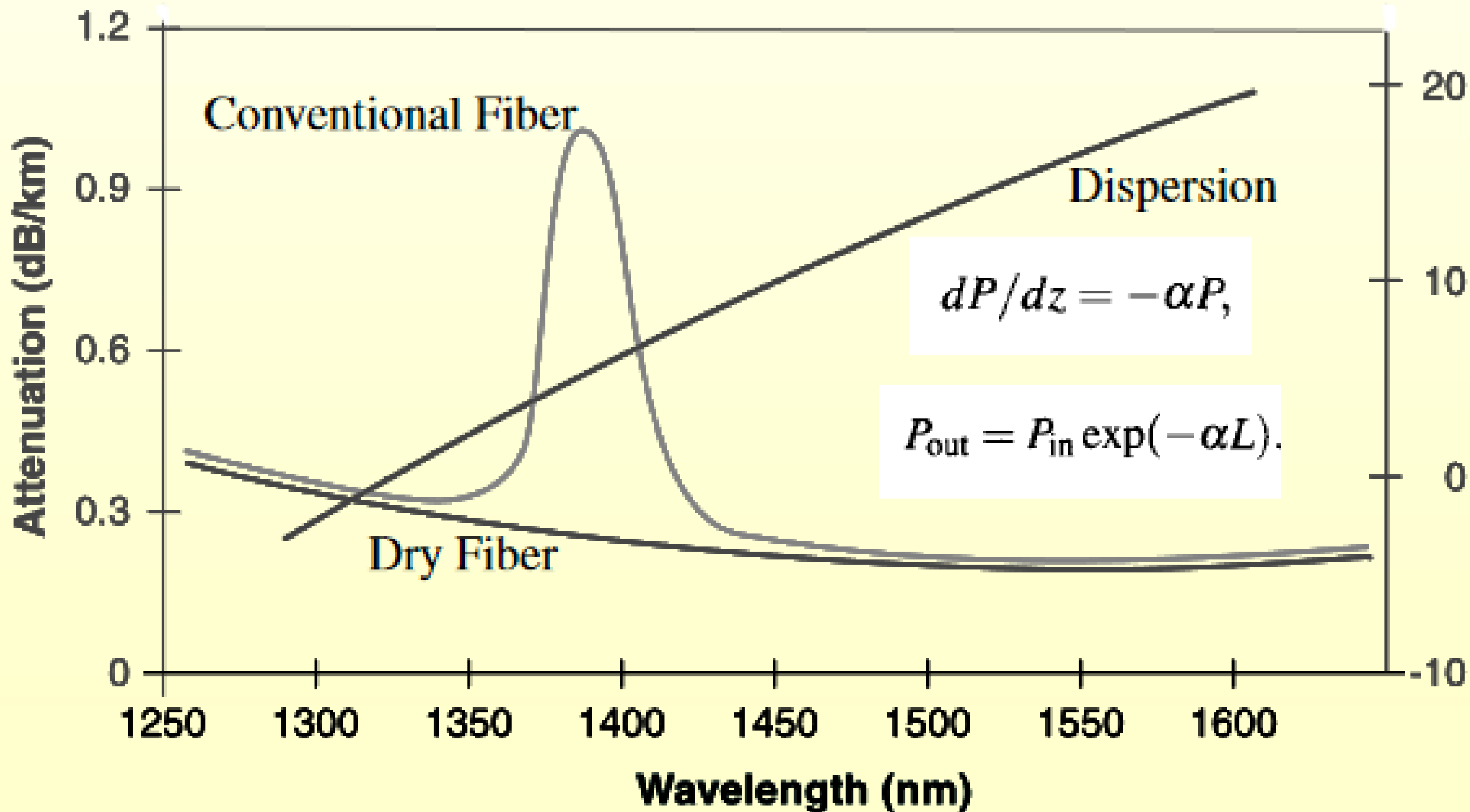


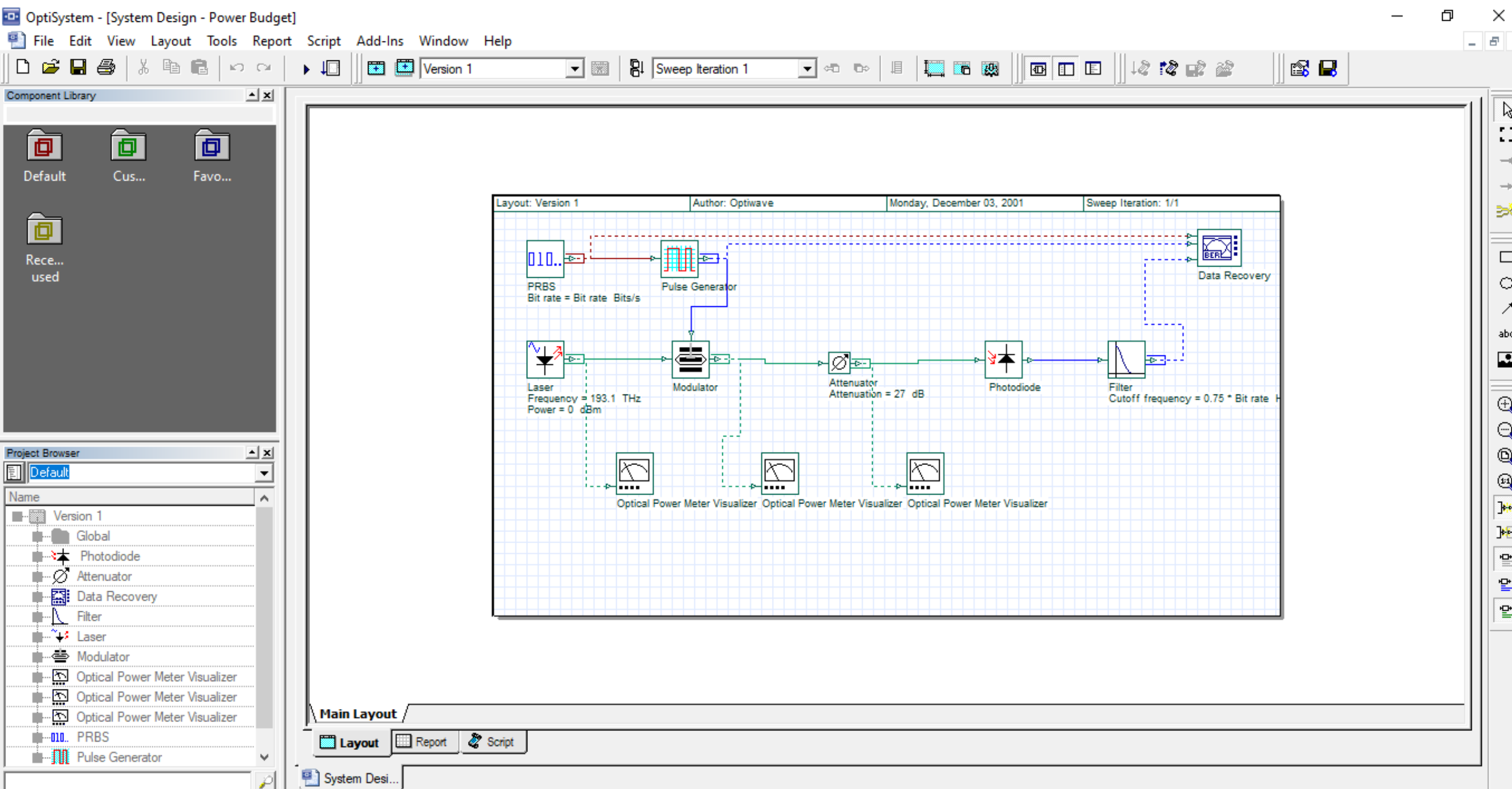
Lightwave System Components

Generic System



- **Optical Transmitters:**
Convert electrical data into an optical bit stream suitable for transmission.
- **Communication Channel:**
Optical fibers are used for transmitting optical bit streams in most terrestrial networks.
- **Optical Receivers:**
Convert optical bit stream into the original electrical form.





Optical Fiber Properties

Optical Fiber Properties

Label: Optical Fiber Cost\$: 0.00

OK Cancel Evaluate Script

Main Disp... PMD Nonl... Num... Gr... Simu... Noise Rand...

Disp	Name	Value	Units	Mode
☐	User defined reference w	☒		Normal
☐	Reference wavelength	1550	nm	Normal
☒	Length	50	km	Normal
☐	Attenuation effect	☒		Normal
☐	Attenuation data type	Constant		Normal
☐	Attenuation	0.2	dB/km	Normal
☐	Attenuation vs. wavelengt	Attenuation.dat		Normal

Load... Save As... Security... Help

Optical Fiber Properties

Label: Optical Fiber Cost\$: 0.00

OK Cancel Evaluate Script

Main Disp... PMD Nonl... Num... Gr... Simu... Noise Rand...

Disp	Name	Value	Units	Mode
☐	Group velocity dispersion	☒		Normal
☐	Third-order dispersion	☒		Normal
☐	Dispersion data type	Constant		Normal
☐	Frequency domain param	☐		Normal
☐	Dispersion	16.75	ps/nm/km	Normal
☐	Dispersion slope	0.075	ps/nm ² /k	Normal
☐	Beta 2	-20	ps ² /km	Normal
☐	Beta 3	0	ps ³ /km	Normal
☐	Dispersion file format	Dispersion vs. wavelength		Normal
☐	Dispersion file name	Dispersion.dat		Normal

Load... Save As... Security... Help

Fiber Specifications

Table 2.1 Characteristics of several commercial fibers

Fiber Type and Trade Name	A_{eff} (μm^2)	λ_{ZD} (nm)	D (C band) [ps/(km-nm)]	Slope S [ps/(km-nm ²)]
Corning SMF-28	80	1302–1322	16 to 19	0.090
OFS AllWave	80	1300–1322	17 to 20	0.088
Draka ColorLock	80	1300–1320	16 to 19	0.090
Corning Vascade	100	1300–1310	18 to 20	0.060
OFS TrueWave-RS	50	1470–1490	2.6 to 6	0.050
Corning LEAF	72	1490–1500	2.0 to 6	0.060
Draka TeraLight	65	1430–1440	5.5 to 10	0.052